1.6 Planes in 3-Space_Guide

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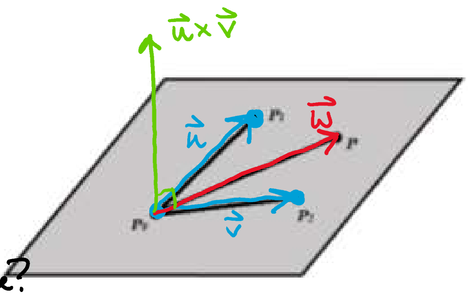
in 3-Spac...

1.6 Planes in 3-Space

We now move to thinking about planes, and we ask ourselves what information do we need in order to determine an equation of a plane?

Suppose we have a plane determined by P_0, P_1, & P_2

If we have another point P (x,y,z) can we determine if P is in the plane?



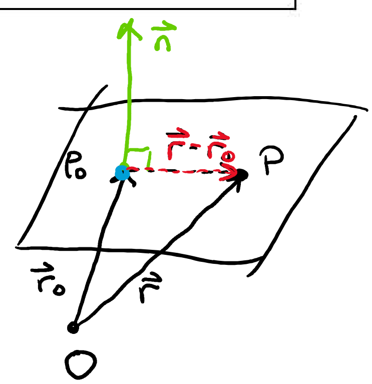
We want to check if $\vec{w}$ is perpendicular to a vector that is perpendicular to the plane (like $\vec{u}\times\vec{v}$)

So we want $(\vec{u}\times\vec{v})\cdot\vec{w}=0$ OR $\vec{n}\cdot\vec{w}=0$

Definition: We say that a vector is normal to a plane if it perpendicular to the plane.

We want to have

$\vec{n}\cdot(\vec{r}-\vec{r}_0)=0$ to have the equation of a plane.



Terminology: If $\mathbf{r}_0$ is the position vector of a point on a plane and $\mathbf{n}$ is a vector normal to the plane, then $\mathbf{n}\cdot(\mathbf{r}-\mathbf{r}_0)=0$ is called the <u>vector equation of the plane</u>.

Note: The vector equation of a plane can be used beyond three dimensions, and it can be used in two dimensions to yield the equation of a line! In this class, planes in four or more dimensions will be known as hyperplanes.

If we return our focus to 3-space and suppose that $\vec{n}=\langle a,b,c\rangle$ $\vec{r}_0=\langle x_0,y_0,z_0\rangle$ $\vec{r}=\langle x,y,z\rangle$

$$\vec{n}\cdot(\vec{r}-\vec{r}_0)=0 \rightarrow \langle a,b,c\rangle\cdot\langle x-x_0,y-y_0,z-z_0\rangle=0$$

$$\rightarrow a(x-x_0)+b(y-y_0)+c(z-z_0)=0$$

Terminology: The equation $a(x-x_0)+b(y-y_0)+c(z-z_0)=0$ is known as the <u>point-normal form</u> of the equation of a plane through (x_0,y_0,z_0) with normal vector $\mathbf{n}=\langle a,b,c\rangle$.

If we let $d=ax_0+by_0+cz_0$

then the equation the plane looks like $ax+by+cz=d$

Theorem: If a, b, c, and d are constants (with a, b, and c not *all* zero), then the graph of the equation

$$ax+by+cz=d$$

is a plane that has the vector $\mathbf{n}=\langle a,b,c\rangle$ as a normal vector.

Note: Alternative approaches to deriving this equation are shown in the notes in Blackboard.

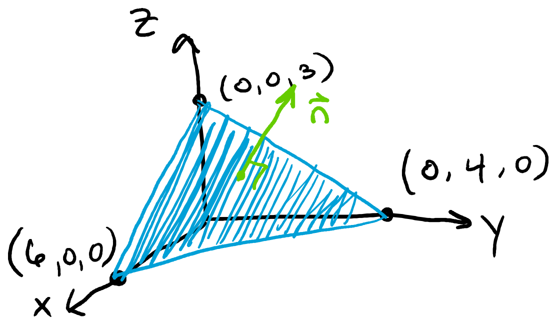
Example 1: Find an equation (in the form $ax+by+cz=d$) of the plane through the point (2,4,-1) with normal vector $\langle 2,3,4\rangle$. Also sketch the portion of this plane which is in the first octant.

$\vec{n} = \langle 2,3,4\rangle$ Point: $(2,4,-1)$

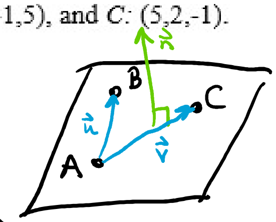
$$2(x-2)+3(y-4)+4(z-(-1))=0$$

$$2x-4+3y-12+4z+4=0$$

$$2x+3y+4z=12$$



Example 2: Find an equation of the plane that passes through the points A: (1,3,3), B: (3,-1,5), and C: (5,2,-1).



Let $\vec{u} = \langle 2,-4,2\rangle$ & $\vec{v} = \langle 4,-1,-4\rangle$

$$\vec{n} = \vec{u}\times\vec{v} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & -4 & 2 \\ 4 & -1 & -4 \end{vmatrix} = \langle 18, -(-16), 14\rangle$$

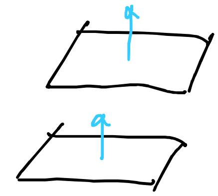
$$18(x-1)+16(y-3)+14(z-3)=0$$

$$18x+16y+14z=108$$

Definition: Two planes are <u>parallel</u> if their normal vectors are parallel. If two planes are not parallel, then they intersect in a straight line and the <u>angle between the two planes</u> is defined to be the acute or right angle between their normal vectors (one must choose the normal vectors so that their dot product is nonnegative).

<u>Example 3:</u> Consider the planes given by $\underbrace{2x-2y+z=4}_{\text{Plane 1}}$ and $\underbrace{x+4y+z=1}_{\text{Plane 2}}$.

a) Determine whether the planes parallel.



So we see $\vec{n}_1 = \langle 2,-2,1 \rangle$ & $\vec{n}_2 = \langle 1,4,1 \rangle$

These are <u>NOT</u> scalar multiples & so $\vec{n}_1$ & $\vec{n}_2$ are not parallel.

Thus Plane 1 & Plane 2 are <u>NOT</u> parallel.

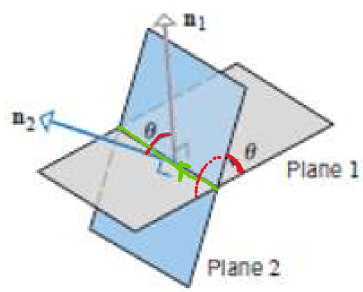
b) Find the angle (rounded to the nearest hundredth) between the planes.

Recall $\vec{a}\cdot\vec{b} = \|\vec{a}\|\cdot\|\vec{b}\|\cos(\theta)$

So $\vec{n}_1\cdot\vec{n}_2 = \|\vec{n}_1\|\cdot\|\vec{n}_2\|\cos(\theta)$

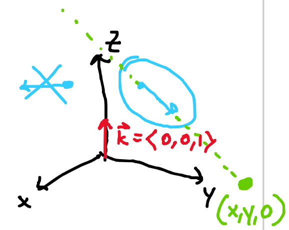
$$\cos(\theta) = \frac{|\vec{n}_1\cdot\vec{n}_2|}{\|\vec{n}_1\|\cdot\|\vec{n}_2\|} = \frac{|2-8+1|}{\sqrt{9}\cdot\sqrt{18}} = \frac{|-5|}{3\sqrt{18}}$$

$$\theta = \cos^{-1}\left(\frac{5}{3\sqrt{18}}\right) \approx 1.17 \approx 66.9^\circ$$



c) Write a parametric equation for the line of intersection for the planes.

We need Point: $\left(\frac{9}{5}, -\frac{1}{5}, 0\right)$ & direction vector: $\vec{v} = \langle -6, -1, 10\rangle$



Let $\vec{v} = \vec{n}_1 \times \vec{n}_2 = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & -2 & 1 \\ 1 & 4 & 1 \end{vmatrix} = \langle -6, -1, 10\rangle$

We check that $\vec{v}\cdot\vec{k} = 0+0+10 = 10 \neq 0$ so $\vec{v}$ & $\vec{k}$ are NOT perpendicular

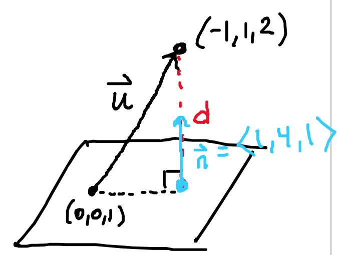
So, our line has a point of the form $(x,y,0)$ & this point must also be on both plane

$$\begin{cases} 2x-2y = 4 \\ x+4y = 1 \end{cases} \longrightarrow \begin{array}{r} 4x+-4y=8 \\ + \quad x+4y=1 \\ \hline 5x=9 \\ x=9/5 \end{array} \qquad \begin{array}{l} \frac{9}{5}+4y=1 \\ 4y=-\frac{4}{5} \\ y=-1/5 \end{array}$$

So $$\boxed{\begin{cases} x = -6t+9/5 \\ y = -1t-1/5 \\ z = 10t+0 \end{cases}}$$

Example 4: Find the exact distance between the point $A:(-1,1,2)$ and the plane given by: $x+4y+z=1$.

We note that $(0, 0, 1)$ is in the plane.



Let $\vec{u} = \langle -1, 1, 1\rangle$

So $d = \left|\text{comp}_{\vec{n}}(\vec{u})\right| = \frac{|\vec{u}\cdot\vec{n}|}{\|\vec{n}\|}$

$$= \frac{-1+4+1}{\sqrt{18}} = \boxed{\frac{4}{\sqrt{18}}}$$

Note: A way of generalizing this process is shown in the notes on Blackboard.